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United States Patent	12412446
Kind Code	B2
Date of Patent	September 09, 2025
Inventor(s)	Small; David et al.

Internal gaming features at gaming devices with transparent displays

Abstract

A gaming device includes a housing defining an interior volume and a transparent display that includes an emissive layer including an array of pixels, each including an emissive pixel element and a transparent portion, and a conductive layer operatively to selectively operate the emissive pixel elements. The gaming device further include a processor circuit and a memory storing machine-readable instructions that, when executed by the processor circuit, cause the processor circuit to selectively operate the transparent display. The transparent portion of each pixel is transparent to visible light in an inactive state such the interior volume of the housing is visible through the transparent portion to a user outside the housing. Each emissive pixel of the array of emissive pixels emits light in an active state sufficient to obscure visibility of the interior volume of the housing through the transparent portion of the pixel to the user outside the housing.

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Appl. No.: **18/087418**

Filed: **December 22, 2022**

Prior Publication Data

Document Identifier	Publication Date
US 20240212423 A1	Jun. 27, 2024

Publication Classification

Int. Cl.: **G07F17/32** (20060101)

U.S. Cl.:

CPC **G07F17/3216** (20130101);

Field of Classification Search

USPC: None

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Background/Summary

BACKGROUND

(1) Embodiments described herein relate to features of electronic wagering devices, and in particular to a gaming device with a transparent display in a gaming environment, such as in a casino environment, and related devices, systems, and methods. Conventional wagering games, such as wagering games provided at Electronic Gaming Machines (EGMs) in a casino environment, may have mechanical elements, such as mechanical slot reels, and may also, or alternatively, include display devices for displaying digital content. There is a need for alternative gaming device arrangements that allow for a player to view and interact with both digital and mechanical elements at the same time and in a more integrated way.

BRIEF SUMMARY

(2) According to some embodiments, a gaming device includes a housing defining an interior volume and a transparent display disposed on the housing. The transparent display includes an emissive layer including an array of pixels, each pixel of the array of pixels including an emissive pixel element and a transparent portion, and a conductive layer operatively coupled to the emissive layer to selectively operate the emissive pixel elements. The gaming device further include a processor circuit and a memory storing machine-readable instructions that, when executed by the processor circuit, cause the processor circuit to selectively operate the transparent display. The transparent portion of each pixel is transparent to visible light in an inactive state such the interior volume of the housing is visible through the transparent portion to a user outside the housing. Each emissive pixel of the array of emissive pixels emits light in an active state sufficient to obscure visibility of the interior volume of the housing through the transparent portion of the pixel to the

user outside the housing.
(3) According to some embodiments a system includes a processor circuit and a memory storing machine-readable instructions that, when executed by the processor circuit, cause the processor circuit to selectively operate a transparent display of a gaming device. The transparent display includes an emissive layer including an array of pixels, each pixel of the array of pixels including an emissive pixel element and a transparent portion, and a conductive layer operatively coupled to the emissive layer to selectively operate the emissive pixel elements. The transparent portion of each pixel is transparent to visible light in an inactive state such an interior volume of the gaming device housing is visible through the transparent portion to a user outside the housing. Each emissive pixel of the array of emissive pixels emits light in an active state sufficient to obscure visibility of the interior volume of the housing through the transparent portion of the pixel to the user outside the housing.
(4) According to some embodiments, a transparent display for a gaming device includes an emissive layer including an array of pixels, each pixel of the array of pixels including an emissive pixel element and a transparent portion, and a conductive layer operatively coupled to the emissive layer to selectively operate the emissive pixel elements. The transparent portion of each pixel is transparent to visible light in an inactive state such an interior volume of a housing of the gaming device is visible through the transparent portion to a user outside the housing. Each emissive pixel of the array of emissive pixels emits light in an active state sufficient to obscure visibility of the interior volume of the housing through the transparent portion of the pixel to the user outside the housing.

Description

BRIEF DESCRIPTION OF SEVERAL VIEWS OF THE DRAWINGS
(1) FIG. **1** is a schematic block diagram illustrating a network configuration for a plurality of gaming devices according to some embodiments.
(2) FIG. **2**A is a perspective view of a gaming device that can be configured according to some embodiments.
(3) FIG. **2**B is a schematic block diagram illustrating an electronic configuration for a gaming device according to some embodiments.
(4) FIG. **2**C is a schematic block diagram that illustrates various functional modules of a gaming device according to some embodiments.
(5) FIG. **2**D is perspective view of a gaming device that can be configured according to some embodiments.
(6) FIG. **2**E is a perspective view of a gaming device according to further embodiments.
(7) FIG. **3**A-**3**C illustrate a gaming device having a transparent display with pixels that allow for selectively facilitating and/or obscuring visibility of internal components behind the display, according to some embodiments.
(8) FIGS. **4**A-**4**D illustrate selectively facilitating and/or obscuring visibility of components through a transparent display of a gaming device, according to some embodiments.
(9) FIGS. **5**A-**5**C illustrates selectively facilitating and/or obscuring visibility of objects through a transparent display of a gaming device, according to some embodiments.
(10) FIGS. **6**A-**6**B illustrates selectively facilitating and/or obscuring visibility of additional internal components through the displays, according to some embodiments.
DETAILED DESCRIPTION
(11) Embodiments described herein relate to features of electronic wagering devices, and in particular to a gaming device with a transparent display in a gaming environment, such as in a casino environment, and related devices, systems, and methods.

(12) According to some embodiments, a gaming device may include a housing defining an interior volume and a transparent display disposed on the housing. The transparent display may include an emissive layer including an array of pixels, each pixel of the array of pixels including an emissive pixel element and a transparent portion, and a conductive layer operatively coupled to the emissive layer to selectively operate the emissive pixel elements. The gaming device may further include a processor circuit and a memory storing machine-readable instructions that, when executed by the processor circuit, cause the processor circuit to selectively operate the transparent display. The transparent portion of each pixel may be transparent to visible light in an inactive state such the interior volume of the housing is visible through the transparent portion to a user outside the housing. Each emissive pixel of the array of emissive pixels may emit light in an active state sufficient to obscure visibility of the interior volume of the housing through the transparent portion of the pixel to the user outside the housing.
(13) Screens in casino environments have become increasingly large over the last years, with some being over 65 inches. This applies to EGMs, ETGs but also for merchandising and infotainment in the casino environment. Although these larger screens provides a large canvas to display content, it can be challenging for a consumer, being near to such a screen, to not get overwhelmed or lost on the casino floor but to remain aware of other offerings across the casino floor. Another downside of these large screens is that the larger a screen is the more difficult is it to have a direct line of sight to other casino offerings.
(14) Many vendors have attempted to address these issues by placing screens in the upper area of the venue, close to the ceiling. This helps to make visitors be more aware of ground-level offerings, but to a drawback on casino surveillance systems, as due to that the cameras' line of sight is majorly impacted.
(15) Transparent displays disclosed herein may be included in EGMs, table games, and other devices and systems, to permit a viewer to look through the screen and actively shift their focus towards content behind the screen, e.g., environmental content, internal components, etc. For example, a viewer could look through an EGM screen to view an environment behind the EGM and/or internal components of the EGM. A viewer could also view an environment behind a virtual table game dealer or view table game components, e.g., a roulette wheel or playing cards. These and other gaming offerings may be associated with events in the casino or other venue to adapt full or partial screen transparency dynamically to the events in real-time. These and other novel technical solution address the technical problem of providing increased player enjoyment and engagement.
(16) These and other embodiments may employ Transparent Organic Light Emitting Diode (T-OLED) screens, which have the ability of dynamically changing certain areas of their screen surface between transparent, semi-transparent, and opaque states. As a result, screens and/or areas thereof can turn transparent when needed to enable a direct line of sight, opaque to have full focus on the content, or semi-transparent to provide a hybrid effect, as desired.
(17) These and other types of transparent displays may also be controlled by a system controller which may be aware of events happening on the casino floor in real time, such as live events, announcements, bingo drawings, jackpot triggers on other gaming devices, happy hour information, etc. By dynamically turning screen contents transparent or semi-transparent the system controller can enable innovative methods of increasing casino visitors' awareness of nearby casino offerings without impacting their casino gaming experience. These and other features can also improve the experience of non-players who can see through the screens and get a more immersive view of the casino interior.
(18) Transparent displays may be included in any number of devices and systems, including EGM base game screens, EGM top screens, screens placed in front of mechanical slot reels, EGM video toppers, etc. Other examples include a virtual dealer screen at a virtual table game (e.g., virtual poker table), screens covering a physical roulette wheel, vertical screens providing additional

information and/or duplicating table game information, individual player screens, and/or a centrally placed screen or "video wall" associated with multiple machines. Optical sensors such as presence detectors, distance meters, gaze trackers, body/head posture trackers, etc., may also be used with the embodiments herein, and the screens may be controlled locally at the individual device, and/or by a central system controller, as desired. Advantages of these and other embodiments include the ability to provide novel form factors and designs, novel gaming features, providing benefits to operators by maintaining line of sight surveillance and players by allowing for different levels of immersion and allowing for greater ease of navigation in a casino environment.

(19) T-OLED displays differ from conventional LED displays in that the pixels in the T-OLED display are self-illuminating, which can eliminate the need for a separate backlight for the display. When combined with a transparent substrate, e.g., glass, portions of the T-OLED display may be opaque when fully illuminated and transparent when deactivated, allowing for portions of the display to be selectively activated and deactivated independently to make the different portions transparent and opaque as desired, in different combinations, and dynamically modified in real time. Many T-OLED displays can be integrated with touch-sensitive sensors without impacting transparency, and may have a very wide field of view approaching 180 degrees.

(20) In some embodiments, a part of the screen may gradually become more transparent and gradually reveal a prize or hidden bonus as the game progresses or as the player increases their bet. EGM features may also be combined with traditional table game features (e.g., mechanical objects such as dice or playing cards) using transparent displays. Integrating physical objects into the game play may increase player engagement and increase player trust and confidence in game results over purely electronically generated game results.

(21) Some embodiments may include a screen-based primary game combined with a bonus or secondary game employing physical objects. For example, the primary game may behave like a conventional slot game, but when entering the dice-based bonus game, some or all of the transparent display may turn fully transparent. A mechanical dice share located behind the transparent display may mechanically roll dice, e.g., by throwing or inducing movement via vibration. Cameras may detect the results of the dice, which may have an impact on the bonus game. In some embodiments, physical playing cards may be employed in card-based games such as blackjack or poker. A blower-type device that pick random balls for a lottery game, or any number of other types of devices, may be employed, as desired.

(22) In some examples, a win presentation may employ the transparent display, such as combining opaque or semi-transparent visual effects on the display with mechanical objects visible through the other portions of the display.

(23) In another embodiment, a transparent display may be placed in front of mechanical EGM game elements, such as a wheel or stepper reels for example, and may dynamically display opaque or semi-transparent content over the mechanical elements. The display(s) may be stationary, with the mechanical elements moving independently of the display. For example, transparent display may be a curved overlay in front of the reels and conforming to the curvature of the reels to minimize the gap in between the reels and OLED display and minimize parallax effects with respect to the display and the reels. In other embodiments, the display(s) may be coupled to the mechanical elements. For example, each symbol on a mechanical reel may include a printed symbol covered by a transparent display, which can be dynamically activated and deactivated to display different symbols or animations during game play, to indicate a win, or to indicate a bonus game or condition, etc. In some examples, the reel may be entirely transparent to view inside the reels, with the symbols being selectively provided by the integrated transparent display(s). This and other embodiments may also enable a player to see inside the stepper reel and view the reel strips from all sides.

(24) The transparent display may also enable viewing of other internal components, for example, by a technician or operator. For example, a technician can view a paper fill level of a ticket printer

through a transparent display during a maintenance operation, with the transparent display being opaque during normal game play operation. In another example, an internal coin hopper can be made visible in an attract mode to allow players to see how many coins are in the hopper. The EGM cabinet could be configured so that the hopper is directly behind the screen, with the hopper components such as the coin bowl being made of transparent plastic. In some examples, non-monetary prizes such as merchandise, keys to a vehicle, etc., can be displayed during game play and/or in an attract mode, such that the display and dispensing process may be made visible to the player.
(25) In some embodiments, mechanical elements of a table game may be combined with or enhanced by a transparent display for providing additional game dynamics, mechanics, and/or other features. For example, a transparent display may be provided on, over, or in front of a mechanical roulette wheel to display additional information, graphical elements such as animations, and/or additional betting options. In some examples, visual effects, animations, win celebrations, etc. may move between a transparent display associated with the roulette wheel and a player's betting screen. The transparent display may be any shape or configuration, such as a square, rectangle, circle, oval, ring, or any other shape, to conform to mechanical elements of the game. In some examples, the transparent display may be transparent or semi-transparent while the roulette wheel is turning, and may become opaque when the ball has landed on a particular number. The transparent display may also supplement or replace an electronic table game (ETG) top screen, enabling viewers to see both relevant game information of the ETG and also an environment behind the display, e.g., a casino environment.
(26) A central controller unit may be used as a casino-wide screen transparency controller, e.g., connected to some or all of the gaming devices, and may be capable of adjusting their level of transparency individually and/or globally. The controller may be aware of any number of device parameters, such as a device type (EGM, ETG, stepper, dual player machine, . . .), a configuration (cabinet type, bank configuration vs. single configuration, type of bank configuration), a number of other devices surrounded by (and their distance to each other), a number of screens, size and transparency capabilities (single screen, dual screen, curved screen, giant screen, . . .), a location (absolute location in the venue, and relative location to other screens/events/amenities), alignment (direction of the player looking at when playing, relative to other screens/events/amenities), awareness about visibility of other machines/casino offerings for the player when screen is turned transparent vs. when screen is turned opaque, occupancy status (player actively playing, player sitting in front, device last time played), a number of detected observers around it (e.g., detected via camera; also, their gaze direction/head posture relative to device), revenue creation information ("hot machine", rarely played machine, above/below floor average), active games installed (multi game vs. single game, no. of games, current game, type of games, themes, . . .), game configurations installed (denom, bet levels, activated features, RTP, . . .), jackpot information (level of jackpot, potential high wins and their value, pulse to hit frequency, . . .), a current transparency level (of each screen, each content type, etc.), actual events/offerings/amenities going on in the casino (concerts, live events, bingo drawings, jackpot triggers, happy hours, casino-wide bonus offerings, . . .), date and time, casino occupancy level, and other device-related information, as desired.
(27) In this regard, examples of systems and gaming devices for a gaming environment will be generally described. In this regard, FIG. **1** illustrates a gaming system **10** including a plurality of gaming devices **100**. As discussed above, the gaming devices **100** may be one type of a variety of different types of gaming devices, such as electronic gaming machines (EGMs), mobile gaming devices, or other devices, for example. The gaming system **10** may be located, for example, on the premises of a gaming establishment, such as a casino. The gaming devices **100**, which are typically situated on a casino floor, may be in communication with each other and/or at least one central controller **40** through a data communication network **50** that may include a remote communication

link. The data communication network **50** may be a private data communication network that is operated, for example, by the gaming facility that operates the gaming devices **100**. Communications over the data communication network **50** may be encrypted for security. The central controller **40** may be any suitable server or computing device which includes at least one processing circuit and at least one memory or storage device. Each gaming device **100** may include a processing circuit that transmits and receives events, messages, commands or any other suitable data or signal between the gaming device **100** and the central controller **40**. The gaming device processing circuit is operable to execute such communicated events, messages or commands in conjunction with the operation of the gaming device **100**. Moreover, the processing circuit of the central controller **40** is configured to transmit and receive events, messages, commands or any other suitable data or signal between the central controller **40** and each of the individual gaming devices **100**. In some embodiments, one or more of the functions of the central controller **40** may be performed by one or more gaming device processing circuits. Moreover, in some embodiments, one or more of the functions of one or more gaming device processing circuits as disclosed herein may be performed by the central controller **40**.

(28) A wireless access point **60** provides wireless access to the data communication network **50**. The wireless access point **60** may be connected to the data communication network **50** as illustrated in FIG. **1**, and/or may be connected directly to the central controller **40** or another server connected to the data communication network **50**.

(29) A player tracking server **45** may also be connected through the data communication network **50**. The player tracking server **45** may manage a player tracking account that tracks the player's gameplay and spending and/or other player preferences and customizations, manages loyalty awards for the player, manages funds deposited or advanced on behalf of the player, and other functions. Player information managed by the player tracking server **45** may be stored in a player information database **47**.

(30) As further illustrated in FIG. **1**, the gaming system **10** may include a ticket server **90** that is configured to print and/or dispense wagering tickets. The ticket server **90** may be in communication with the central controller **40** through the data communication network **50**. Each ticket server **90** may include a processing circuit that transmits and receives events, messages, commands or any other suitable data or signal between the ticket server **90** and the central controller **40**. The ticket server **90** processing circuit may be operable to execute such communicated events, messages or commands in conjunction with the operation of the ticket server **90**. Moreover, in some embodiments, one or more of the functions of one or more ticket server **90** processing circuits as disclosed herein may be performed by the central controller **40**.

(31) The gaming devices **100** communicate with one or more elements of the gaming system **10** to coordinate providing wagering games and other functionality. For example, in some embodiments, the gaming device **100** may communicate directly with the ticket server **90** over a wireless interface **62**, which may be a WiFi link, a Bluetooth link, a near field communications (NFC) link, etc. In other embodiments, the gaming device **100** may communicate with the data communication network **50** (and devices connected thereto, including other gaming devices **100**) over a wireless interface **64** with the wireless access point **60**. The wireless interface **64** may include a WiFi link, a Bluetooth link, an NFC link, etc. In still further embodiments, the gaming devices **100** may communicate simultaneously with both the ticket server **90** over the wireless interface **66** and the wireless access point **60** over the wireless interface **64**. Some embodiments provide that gaming devices **100** may communicate with other gaming devices over a wireless interface **64**. In these embodiments, wireless interface **62**, wireless interface **64** and wireless interface **66** may use different communication protocols and/or different communication resources, such as different frequencies, time slots, spreading codes, etc.

(32) Embodiments herein may include different types of gaming devices. One example of a gaming device includes a gaming device **100** that can use gesture and/or touch-based inputs according to

various embodiments is illustrated in FIGS. **2**A, **2**B, and **2**C in which FIG. **2**A is a perspective view of a gaming device **100** illustrating various physical features of the device, FIG. **2**B is a functional block diagram that schematically illustrates an electronic relationship of various elements of the gaming device **100**, and FIG. **2**C illustrates various functional modules that can be stored in a memory device of the gaming device **100**. The embodiments shown in FIGS. **2**A to **2**C are provided as examples for illustrative purposes only. It will be appreciated that gaming devices may come in many different shapes, sizes, layouts, form factors, and configurations, and with varying numbers and types of input and output devices, and that embodiments are not limited to the particular gaming device structures described herein.

(33) Gaming devices **100** typically include a number of standard features, many of which are illustrated in FIGS. **2**A and **2**B. For example, referring to FIG. **2**A, a gaming device **100** (which is an EGM **160** in this embodiment) may include a support structure, housing **105** (e.g., cabinet) which provides support for a plurality of displays, inputs, outputs, controls and other features that enable a player to interact with the gaming device **100**.

(34) The gaming device **100** illustrated in FIG. **2**A includes a number of display devices, including a primary display device **116** located in a central portion of the housing **105** and a secondary display device **118** located in an upper portion of the housing **105**. A plurality of game components **155** are displayed on a display screen **117** of the primary display device **116**. It will be appreciated that one or more of the display devices **116**, **118** may be omitted, or that the display devices **116**, **118** may be combined into a single display device. The gaming device **100** may further include a player tracking display **142**, a credit display **120**, and a bet display **122**. The credit display **120** displays a player's current number of credits, cash, account balance or the equivalent. The bet display **122** displays a player's amount wagered. Locations of these displays are merely illustrative as any of these displays may be located anywhere on the gaming device **100**.

(35) The player tracking display **142** may be used to display a service window that allows the player to interact with, for example, their player loyalty account to obtain features, bonuses, comps, etc. In other embodiments, additional display screens may be provided beyond those illustrated in FIG. **2**A. In some embodiments, one or more of the player tracking display **142**, the credit display **120** and the bet display **122** may be displayed in one or more portions of one or more other displays that display other game related visual content. For example, one or more of the player tracking display **142**, the credit display **120** and the bet display **122** may be displayed in a picture in a picture on one or more displays.

(36) The gaming device **100** may further include a number of input devices **130** that allow a player to provide various inputs to the gaming device **100**, either before, during or after a game has been played. The gaming device may further include a game play initiation button **132** and a cashout button **134**. The cashout button **134** is utilized to receive a cash payment or any other suitable form of payment corresponding to a quantity of remaining credits of a credit display.

(37) In some embodiments, one or more input devices of the gaming device **100** are one or more game play activation devices that are each used to initiate a play of a game on the gaming device **100** or a sequence of events associated with the gaming device **100** following appropriate funding of the gaming device **100**. The example gaming device **100** illustrated in FIG. **2**A and **2**B includes a game play activation device in the form of a game play initiation button **132**. It should be appreciated that, in other embodiments, the gaming device **100** begins game play automatically upon appropriate funding rather than upon utilization of the game play activation device.

(38) In some embodiments, one or more input device **130** of the gaming device **100** may include wagering or betting functionality. For example, a maximum wagering or betting function may be provided that, when utilized, causes a maximum wager to be placed. Another such wagering or betting function is a repeat the bet device that, when utilized, causes the previously-placed wager to be placed. A further such wagering or betting function is a bet one function. A bet is placed upon utilization of the bet one function. The bet is increased by one credit each time the bet one device is

utilized. Upon the utilization of the bet one function, a quantity of credits shown in a credit display (as described below) decreases by one, and a number of credits shown in a bet display (as described below) increases by one.
(39) In some embodiments, as shown in FIG. **2**B, the input device(s) **130** may include and/or interact with additional components, such as gesture sensors **156** for gesture input devices, and/or a touch-sensitive display that includes a digitizer **152** and a touchscreen controller **154** for touch input devices, as disclosed herein. The player may interact with the gaming device **100** by touching virtual buttons on one or more of the display devices **116**, **118**, **140**. Accordingly, any of the above-described input devices, such as the input device **130**, the game play initiation button **132** and/or the cashout button **134** may be provided as virtual buttons or regions on one or more of the display devices **116**, **118**, **140**.
(40) Referring briefly to FIG. **2**B, operation of the primary display device **116**, the secondary display device **118** and the player tracking display **142** may be controlled by a video controller **30** that receives video data from a processing circuit **12** or directly from a memory device **14** and displays the video data on the display screen. The credit display **120** and the bet display **122** are typically implemented as simple liquid crystal display (LCD) or light emitting diode (LED) displays that display a number of credits available for wagering and a number of credits being wagered on a particular game. Accordingly, the credit display **120** and the bet display **122** may be driven directly by the processing circuit **12**. In some embodiments however, the credit display **120** and/or the bet display **122** may be driven by the video controller **30**.
(41) Referring again to FIG. **2**A, the display devices **116**, **118**, **140** may include, without limitation: a cathode ray tube, a plasma display, an LCD, a display based on LEDs, a display based on a plurality of organic light-emitting diodes (OLEDs), a display based on polymer light-emitting diodes (PLEDs), a display based on a plurality of surface-conduction electron-emitters (SEDs), a display including a projected and/or reflected image, or any other suitable electronic device or display mechanism. In certain embodiments, as described above, the display devices **116**, **118**, **140** may include a touch-screen with an associated touchscreen controller **154** and digitizer **152**. The display devices **116**, **118**, **140** may be of any suitable size, shape, and/or configuration. The display devices **116**, **118**, **140** may include flat or curved display surfaces.
(42) The display devices **116**, **118**, **140** and video controller **30** of the gaming device **100** are generally configured to display one or more game and/or non-game images, symbols, and indicia. In certain embodiments, the display devices **116**, **118**, **140** of the gaming device **100** are configured to display any suitable visual representation or exhibition of the movement of objects; dynamic lighting; video images; images of people, characters, places, things, and faces of cards; and the like. In certain embodiments, the display devices **116**, **118**, **140** of the gaming device **100** are configured to display one or more virtual reels, one or more virtual wheels, and/or one or more virtual dice. In other embodiments, certain of the displayed images, symbols, and indicia are in mechanical form. That is, in these embodiments, the display device **116**, **118**, **140** includes any electromechanical device, such as one or more rotatable wheels, one or more reels, and/or one or more dice, configured to display at least one or a plurality of game or other suitable images, symbols, or indicia.
(43) The gaming device **100** also includes various features that enable a player to deposit credits in the gaming device **100** and withdraw credits from the gaming device **100**, such as in the form of a payout of winnings, credits, etc. For example, the gaming device **100** may include a bill/ticket dispenser **136**, a bill/ticket acceptor **128**, and a coin acceptor **126** that allows the player to deposit coins into the gaming device **100**.
(44) As illustrated in FIG. **2**A, the gaming device **100** may also include a currency dispenser **137** that may include a note dispenser configured to dispense paper currency and/or a coin generator configured to dispense coins or tokens in a coin payout tray.
(45) The gaming device **100** may further include one or more speakers **150** controlled by one or

more sound cards **28** (FIG. **2**B). The gaming device **100** illustrated in FIG. **2**A includes a pair of speakers **150**. In other embodiments, additional speakers, such as surround sound speakers, may be provided within or on the housing **105**. Moreover, the gaming device **100** may include built-in seating with integrated headrest speakers.

(46) In various embodiments, the gaming device **100** may generate dynamic sounds coupled with attractive multimedia images displayed on one or more of the display devices **116**, **118**, **140** to provide an audio-visual representation or to otherwise display full-motion video with sound to attract players to the gaming device **100** and/or to engage the player during gameplay. In certain embodiments, the gaming device **100** may display a sequence of audio and/or visual attraction messages during idle periods to attract potential players to the gaming device **100**. The videos may be customized to provide any appropriate information.

(47) The gaming device **100** may further include a card reader **138** that is configured to read magnetic stripe cards, such as player loyalty/tracking cards, chip cards, and the like. In some embodiments, a player may insert an identification card into a card reader of the gaming device. In some embodiments, the identification card is a smart card having a programmed microchip or a magnetic strip coded with a player's identification, credit totals (or related data) and other relevant information. In other embodiments, a player may carry a portable device, such as a cell phone, a radio frequency identification tag or any other suitable wireless device, which communicates a player's identification, credit totals (or related data) and other relevant information to the gaming device. In some embodiments, money may be transferred to a gaming device through electronic funds transfer. When a player funds the gaming device, the processing circuit determines the amount of funds entered and displays the corresponding amount on the credit or other suitable display as described above.

(48) In some embodiments, the gaming device **100** may include an electronic payout device or module configured to fund an electronically recordable identification card or smart card or a bank or other account via an electronic funds transfer to or from the gaming device **100**.

(49) FIG. **2**B is a block diagram that illustrates logical and functional relationships between various components of a gaming device **100**. It should also be understood that components described in FIG. **2**B may also be used in other computing devices, as desired, such as mobile computing devices for example. As shown in FIG. **2**B, the gaming device **100** may include a processing circuit **12** that controls operations of the gaming device **100**. Although illustrated as a single processing circuit, multiple special purpose and/or general purpose processors and/or processor cores may be provided in the gaming device **100**. For example, the gaming device **100** may include one or more of a video processor, a signal processor, a sound processor and/or a communication controller that performs one or more control functions within the gaming device **100**. The processing circuit **12** may be variously referred to as a "controller," "microcontroller," "microprocessor" or simply a "computer." The processor may further include one or more application-specific integrated circuits (ASICs).

(50) Various components of the gaming device **100** are illustrated in FIG. **2**B as being connected to the processing circuit **12**. It will be appreciated that the components may be connected to the processing circuit **12** through a system bus **151**, a communication bus and controller, such as a universal serial bus (USB) controller and USB bus, a network interface, or any other suitable type of connection.

(51) The gaming device **100** further includes a memory device **14** that stores one or more functional modules **20**. Various functional modules **20** of the gaming device **100** will be described in more detail below in connection with FIG. **2**D.

(52) The memory device **14** may store program code and instructions, executable by the processing circuit **12**, to control the gaming device **100**. The memory device **14** may also store other data such as image data, event data, player input data, random or pseudo-random number generators, pay-table data or information and applicable game rules that relate to the play of the gaming device.

The memory device **14** may include random access memory (RAM), which can include non-volatile RAM (NVRAM), magnetic RAM (ARAM), ferroelectric RAM (FeRAM) and other forms as commonly understood in the gaming industry. In some embodiments, the memory device **14** may include read only memory (ROM). In some embodiments, the memory device **14** may include flash memory and/or EEPROM (electrically erasable programmable read only memory). Any other suitable magnetic, optical and/or semiconductor memory may operate in conjunction with the gaming device disclosed herein.

(53) The gaming device **100** may further include a data storage **22**, such as a hard disk drive or flash memory. The data storage **22** may store program data, player data, audit trail data or any other type of data. The data storage **22** may include a detachable or removable memory device, including, but not limited to, a suitable cartridge, disk, CD ROM, Digital Video Disc ("DVD") or USB memory device.

(54) The gaming device **100** may include a communication adapter **26** that enables the gaming device **100** to communicate with remote devices over a wired and/or wireless communication network, such as a local area network (LAN), wide area network (WAN), cellular communication network, or other data communication network. The communication adapter **26** may further include circuitry for supporting short range wireless communication protocols, such as Bluetooth and/or NFC that enable the gaming device **100** to communicate, for example, with a mobile communication device operated by a player.

(55) The gaming device **100** may include one or more internal or external communication ports that enable the processing circuit **12** to communicate with and to operate with internal or external peripheral devices, such as eye tracking devices, position tracking devices, cameras, accelerometers, arcade sticks, bar code readers, bill validators, biometric input devices, bonus devices, button panels, card readers, coin dispensers, coin hoppers, display screens or other displays or video sources, expansion buses, information panels, keypads, lights, mass storage devices, microphones, motion sensors, motors, printers, reels, Small Computer System Interface ("SCSI") ports, solenoids, speakers, thumb drives, ticket readers, touch screens, trackballs, touchpads, wheels, and wireless communication devices. In some embodiments, internal or external peripheral devices may communicate with the processing circuit through a USB hub (not shown) connected to the processing circuit **12**.

(56) In some embodiments, the gaming device **100** may include a sensor, such as a camera **127**, in communication with the processing circuit **12** (and possibly controlled by the processing circuit **12**) that is selectively positioned to acquire an image of a player actively using the gaming device **100** and/or the surrounding area of the gaming device **100**. In one embodiment, the camera **127** may be configured to selectively acquire still or moving (e.g., video) images and may be configured to acquire the images in either an analog, digital or other suitable format. The display devices **116**, **118**, **140** may be configured to display the image acquired by the camera **127** as well as display the visible manifestation of the game in split screen or picture-in-picture fashion. For example, the camera **127** may acquire an image of the player and the processing circuit **12** may incorporate that image into the primary and/or secondary game as a game image, symbol or indicia.

(57) Various functional modules of that may be stored in a memory device **14** of a gaming device **100** are illustrated in FIG. **2**C. Referring to FIG. **2**C, the gaming device **100** may include in the memory device **14** a game module **20**A that includes program instructions and/or data for operating a hybrid wagering game as described herein. The gaming device **100** may further include a player tracking module **20**B, an electronic funds transfer module **20**C, an input device interface **20**D, an audit/reporting module **20**E, a communication module **20**F, an operating system kernel **20**G and a random number generator **20**H. The player tracking module **20**B keeps track of the play of a player. The electronic funds transfer module **20**C communicates with a back end server or financial institution to transfer funds to and from an account associated with the player. The input device interface **20**D interacts with input devices, such as the input device **130**, as described in more detail

below. The communication module **20**F enables the gaming device **100** to communicate with remote servers and other gaming devices using various secure communication interfaces. The operating system kernel **20**G controls the overall operation of the gaming device **100**, including the loading and operation of other modules. The random number generator **20**H generates random or pseudorandom numbers for use in the operation of the hybrid games described herein.
(58) In some embodiments, a gaming device **100** includes a personal device, such as a desktop computer, a laptop computer, a mobile device, a tablet computer or computing device, a personal digital assistant (PDA), or other portable computing devices. In some embodiments, the gaming device **100** may be operable over a wireless network, such as part of a wireless gaming system. In such embodiments, the gaming machine may be a hand-held device, a mobile device or any other suitable wireless device that enables a player to play any suitable game at a variety of different locations. It should be appreciated that a gaming device or gaming machine as disclosed herein may be a device that has obtained approval from a regulatory gaming commission or a device that has not obtained approval from a regulatory gaming commission.
(59) For example, referring to FIG. **2**D, a gaming device **100** (which is a mobile gaming device **170** in this embodiment) may be implemented as a handheld device including a compact housing **105** on which is mounted a touchscreen display device **116** including a digitizer **152**. One or more input devices **130** may be included for providing functionality of for embodiments described herein. A camera **127** may be provided in a front face of the housing **105**. The housing **105** may include one or more speakers **150**. In the gaming device **100**, various input buttons described above, such as the cashout button, gameplay activation button, etc., may be implemented as soft buttons on the touchscreen display device **116** and/or input device **130**. In this embodiment, the input device **130** is integrated into the touchscreen display device **116**, but it should be understood that the input device may also, or alternatively, be separate from the display device **116**. Moreover, the gaming device **100** may omit certain features, such as a bill acceptor, a ticket generator, a coin acceptor or dispenser, a card reader, secondary displays, a bet display, a credit display, etc. Credits can be deposited in or transferred from the gaming device **100** electronically.
(60) FIG. **2**E illustrates a standalone gaming device **100** (which is an EGM **160** in this embodiment) having a different form factor from the EGM **160** illustrated in FIG. **2**A. In particular, the gaming device **100** is characterized by having a large, high aspect ratio, curved primary display device **116** provided in the housing **105**, with no secondary display device. The primary display device **116** may include a digitizer **152** to allow touchscreen interaction with the primary display device **116**. The gaming device **100** may further include a player tracking display **142**, an input device **130**, a bill/ticket acceptor **128**, a card reader **138**, and a bill/ticket dispenser **136**. The gaming device **100** may further include one or more cameras **127** to enable facial recognition and/or motion tracking.
(61) Although illustrated as certain gaming devices, such as electronic gaming machines (EGMs) and mobile gaming devices, functions and/or operations as described herein may also include wagering stations that may include electronic game tables, conventional game tables including those involving cards, dice and/or roulette, and/or other wagering stations such as sports book stations, video poker games, skill-based games, virtual casino-style table games, or other casino or non-casino style games. Further, gaming devices according to embodiments herein may be implemented using other computing devices and mobile devices, such as smart phones, tablets, and/or personal computers, among others.
(62) In this regard, FIGS. **3**A-**3**C illustrate a gaming device **300** having a transparent display **304** with pixels **312** that allow for selectively facilitating and/or obscuring visibility of components behind the display **304**, according to some embodiments. As shown by FIG. **3**A, the gaming device **300** in this embodiment includes a housing **302** defining an interior volume **303** and a transparent display **304** disposed on the housing **302**. FIG. **3**B illustrates a cross sectional view of the transparent display **304**, including an emissive layer **306**, a conductive layer **308** operatively

coupled to the emissive layer **306**, and one or more transparent glass layers **310**. In this example, the emissive layer **306** and the conductive layer **308** are attached to the transparent glass layer(s) **310** to provide structural support and protection for the emissive layer **306** and/or the conductive layer **308**.

(63) As shown by FIG. **3**C, the emissive layer **306** includes an array of pixels **312**, with each pixel **312** including an emissive pixel element **314** and a transparent portion **316**. In this example, each emissive pixel element **314** includes an Organic Light Emitting Diode (OLED) element **318**. Unlike conventional liquid crystal displays, OLED elements **318** and other types of emissive pixel elements **314** are self-illuminating, such that the conductive layer **308** may selectively operate the emissive pixel elements **314** individually to emit light in an active state while other pixels **312** may remain unilluminated in an inactive state.

(64) In this example, the transparent portion **316** of each pixel **312** is transparent to visible light in the inactive state such that an interior volume **303** behind the transparent display **304** is visible through the transparent portion **316**. In this example, a relative area of the transparent portion **316** is larger than the area of the emissive pixel element **314**, such that the entire pixel **312** is substantially transparent. Meanwhile, each pixel **312** in an active state emits light sufficient to obscure visibility of the interior volume **303** behind the transparent display **304** through the pixel **312**. For example, the OLED element **318** in this example may self-illuminate with sufficient brightness to overpower and ambient light passing through the transparent portion **316** such that the interior volume **303** is not visible through the transparent portion **316** of the illuminated pixel **312**.

(65) In some embodiments, an area of the transparent portion **316** of each pixel **312** may be between 5% and 95% of a total area of the pixel **312**. For example, as pixel brightness increases and as pixel size is reduced, the transparent portion **316** of the pixel **312** may approach 100% of the total area of the pixel **312**. In some embodiments, the array of pixels **312** may have a pixel density that may surpass the ability of the human eye to detect individual pixels at an appropriate viewing distance. For example, for larger displays at longer viewing distances, pixel densities of 100 dots per inch (DPI) or less may be appropriate. On the other hand, for smaller displays at shorter viewing distances, pixel densities of 2000 dots per inch (DPI) or more may be appropriate.

(66) Referring now to FIGS. **4**A-**4**D, a gaming device **400** similar to the gaming device **300** of FIGS. **3**A-**3**C is provided. In this example, the gaming device **400** may determine that a component **420** in the interior volume **403** of the housing **402** behind the transparent display **404** of the gaming device **400** should be visible to a user **424** of the gaming device **400**. In some examples, the gaming device **400** may employ a gaze tracking sensor **434** to determine a gaze direction of the user **424** selectively operate the transparent display **404** to reduce or increase visibility of specific components **420** in the interior volume **403**.

(67) As shown by FIGS. **4**B and **4**C, the gaming device **400** may selectively operate the transparent display **404** to cause a subset **427** of the array of pixels of the transparent display **404** to be in an active state to obscure visibility of the component **420** to the user **424**. For example, the device **400** may determine that the component **420** (a slot reel **422** in this example) is displaying an unfavorable game symbol **423**, and may position a graphical game element **428** (a more favorable game symbol **425** in this example) on the transparent display **404** such that the game element **428** is positioned in the field of view of the user **424** between the unfavorable symbol **423** and the user **424**.

(68) As shown by FIG. **4**D, the transparent display **404** may have different shapes based on the configuration of the internal components **420**. For example, in this embodiment, the transparent display **404** has a curved portion **430** with a curvature that corresponds to a curvature of the mechanical slot reels **422**, such that a distance **432** between the portion of the transparent display and the plurality of mechanical reels is substantially uniform.

(69) FIGS. **5**A-**5**C illustrate selectively facilitating and/or obscuring visibility of objects **520** through a transparent display **504** of a gaming device **500**, according to some embodiments. For

example, as shown by FIG. 5A, the housing **502** may contain a mechanism that displays and manipulates a plurality of physical dice **538** disposed in the interior volume **503** of the housing **502** for a dice-based game. In this example, the plurality of physical dice **538** is visible to the user outside the housing **502** through the transparent display **504** when the emissive pixels are in the inactive state.

(70) As shown by FIG. 5B, the housing **502** may contain a mechanism that displays and manipulates a plurality of physical playing cards **540** disposed in the interior volume **503** of the housing **502** for a card-based game. In this example, the plurality of physical playing cards **540** is visible to the user outside the housing **502** through the transparent display **504** when the emissive pixels are in the inactive state.

(71) As shown by FIG. 5C, the housing **502** may contain a mechanism that displays a physical non-monetary prize **542** (e.g., a piece of jewelry in this example) disposed in the interior volume **503** of the housing **502** as a prize for the wagering game at the gaming device **500**. In this example, the non-monetary prize **542** is visible to the user outside the housing **502** through the transparent display **504** when the emissive pixels are in the inactive state.

(72) FIGS. **6**A and **6**B illustrate selectively facilitating and/or obscuring visibility of additional internal components **620** through a transparent display **604** of a gaming device **600**, according to some embodiments. As shown by FIG. **6**A the housing **602** may contain a ticket printer device **644** disposed in the interior volume **603** of the housing **602**. The ticket printer device **644** may include a paper receptacle **646** that is visible to a user (e.g., an operator or service technician outside the housing **602** through the plurality of emissive pixels when the emissive pixels are in the inactive state, to facilitate visibility of a paper fill level **648** of the paper receptacle **646**. To further facilitate visibility of a paper fill level **648** of the paper receptacle **646**, the a portion of the paper receptacle **646** may also be transparent. Meanwhile, when the emissive pixels are in an active state, the paper receptacle **646** may not be visible, e.g., to a player at the gaming device **600**.

(73) As shown by FIG. **6**B, the housing **602** may contain a token hopper device **650** disposed in the interior volume **603** of the housing **602**. The token hopper device **650** may include a token receptacle **646** that is visible to a user (e.g., an operator or service technician outside the housing **602** through the plurality of emissive pixels when the emissive pixels are in the inactive state, to facilitate visibility of a token fill level **654** of the token receptacle **652**. To further facilitate visibility of a token fill level **654** of the token receptacle **652**, a portion of the token receptacle **652** may also be transparent Meanwhile, when the emissive pixels are in an active state, the token receptacle **652** may not be visible, e.g., to a player at the gaming device **600**.

(74) Embodiments described herein may be implemented in various configurations for gaming devices **100**, including but not limited to: (1) a dedicated gaming device, wherein the computerized instructions for controlling any games (which are provided by the gaming device) are provided with the gaming device prior to delivery to a gaming establishment; and (2) a changeable gaming device, where the computerized instructions for controlling any games (which are provided by the gaming device) are downloadable to the gaming device through a data network when the gaming device is in a gaming establishment. In some embodiments, the computerized instructions for controlling any games are executed by at least one central server, central controller or remote host. In such a “thin client” embodiment, the central server remotely controls any games (or other suitable interfaces) and the gaming device is utilized to display such games (or suitable interfaces) and receive one or more inputs or commands from a player. In another embodiment, the computerized instructions for controlling any games are communicated from the central server, central controller or remote host to a gaming device local processor and memory devices. In such a “thick client” embodiment, the gaming device local processor executes the communicated computerized instructions to control any games (or other suitable interfaces) provided to a player.

(75) In some embodiments, a gaming device may be operated by a mobile device, such as a mobile telephone, tablet other mobile computing device. For example, a mobile device may be

communicatively coupled to a gaming device and may include a user interface that receives user inputs that are received to control the gaming device. The user inputs may be received by the gaming device via the mobile device.
(76) In some embodiments, one or more gaming devices in a gaming system may be thin client gaming devices and one or more gaming devices in the gaming system may be thick client gaming devices. In another embodiment, certain functions of the gaming device are implemented in a thin client environment and certain other functions of the gaming device are implemented in a thick client environment. In one such embodiment, computerized instructions for controlling any primary games are communicated from the central server to the gaming device in a thick client configuration and computerized instructions for controlling any secondary games or bonus functions are executed by a central server in a thin client configuration.
(77) The present disclosure contemplates a variety of different gaming systems each having one or more of a plurality of different features, attributes, or characteristics. It should be appreciated that a "gaming system" as used herein refers to various configurations of: (a) one or more central servers, central controllers, or remote hosts; (b) one or more gaming devices; and/or (c) one or more personal gaming devices, such as desktop computers, laptop computers, tablet computers or computing devices, PDAs, mobile telephones such as smart phones, and other mobile computing devices.
(78) In certain such embodiments, computerized instructions for controlling any games (such as any primary or base games and/or any secondary or bonus games) displayed by the gaming device are executed by the central server, central controller, or remote host. In such "thin client" embodiments, the central server, central controller, or remote host remotely controls any games (or other suitable interfaces) displayed by the gaming device, and the gaming device is utilized to display such games (or suitable interfaces) and to receive one or more inputs or commands. In other such embodiments, computerized instructions for controlling any games displayed by the gaming device are communicated from the central server, central controller, or remote host to the gaming device and are stored in at least one memory device of the gaming device. In such "thick client" embodiments, the at least one processor of the gaming device executes the computerized instructions to control any games (or other suitable interfaces) displayed by the gaming device.
(79) In some embodiments in which the gaming system includes: (a) a gaming device configured to communicate with a central server, central controller, or remote host through a data network; and/or (b) a plurality of gaming devices configured to communicate with one another through a data network, the data network is an internet or an intranet. In certain such embodiments, an internet browser of the gaming device is usable to access an internet game page from any location where an internet connection is available. In one such embodiment, after the internet game page is accessed, the central server, central controller, or remote host identifies a player prior to enabling that player to place any wagers on any plays of any wagering games. In one example, the central server, central controller, or remote host identifies the player by requiring a player account of the player to be logged into via an input of a unique username and password combination assigned to the player. It should be appreciated, however, that the central server, central controller, or remote host may identify the player in any other suitable manner, such as by validating a player tracking identification number associated with the player; by reading a player tracking card or other smart card inserted into a card reader (as described below); by validating a unique player identification number associated with the player by the central server, central controller, or remote host; or by identifying the gaming device, such as by identifying the MAC address or the IP address of the internet facilitator. In various embodiments, once the central server, central controller, or remote host identifies the player, the central server, central controller, or remote host enables placement of one or more wagers on one or more plays of one or more primary or base games and/or one or more secondary or bonus games, and displays those plays via the internet browser of the gaming device.

(80) It should be appreciated that the central server, central controller, or remote host and the gaming device are configured to connect to the data network or remote communications link in any suitable manner. In various embodiments, such a connection is accomplished via: a conventional phone line or other data transmission line, a digital subscriber line (DSL), a T-1 line, a coaxial cable, a fiber optic cable, a wireless or wired routing device, a mobile communications network connection (such as a cellular network or mobile internet network), or any other suitable medium. It should be appreciated that the expansion in the quantity of computing devices and the quantity and speed of internet connections in recent years increases opportunities for players to use a variety of gaming devices to play games from an ever-increasing quantity of remote sites. It should also be appreciated that the enhanced bandwidth of digital wireless communications may render such technology suitable for some or all communications, particularly if such communications are encrypted. Higher data transmission speeds may be useful for enhancing the sophistication and response of the display and interaction with players.
(81) In the above-description of various embodiments, various aspects may be illustrated and described herein in any of a number of patentable classes or contexts including any new and useful process, machine, manufacture, or composition of matter, or any new and useful improvement thereof. Accordingly, various embodiments described herein may be implemented entirely by hardware, entirely by software (including firmware, resident software, micro-code, etc.) or by combining software and hardware implementation that may all generally be referred to herein as a "circuit," "module," "component," or "system." Furthermore, various embodiments described herein may take the form of a computer program product including one or more computer readable media having computer readable program code embodied thereon.
(82) Any combination of one or more computer readable media may be used. The computer readable media may be a computer readable signal medium or a computer readable storage medium. A computer readable storage medium may be, for example, but not limited to, an electronic, magnetic, optical, electromagnetic, or semiconductor system, apparatus, or device, or any suitable combination of the foregoing. More specific examples (a non-exhaustive list) of the computer readable storage medium would include the following: a portable computer diskette, a hard disk, a random access memory (RAM), a read-only memory (ROM), an erasable programmable read-only memory (EPROM or Flash memory), an appropriate optical fiber with a repeater, a portable compact disc read-only memory (CD-ROM), an optical storage device, a magnetic storage device, or any suitable combination of the foregoing. In the context of this document, a computer readable storage medium may be any medium that can contain, or store a program for use by or in connection with an instruction execution system, apparatus, or device.
(83) A computer readable signal medium may include a propagated data signal with computer readable program code embodied therein, for example, in baseband or as part of a carrier wave. Such a propagated signal may take any of a variety of forms, including, but not limited to, electro-magnetic, optical, or any suitable combination thereof. A computer readable signal medium may be any computer readable medium that is not a computer readable storage medium and that can communicate, propagate, or transport a program for use by or in connection with an instruction execution system, apparatus, or device. Program code embodied on a computer readable signal medium may be transmitted using any appropriate medium, including but not limited to wireless, wireline, optical fiber cable, radio frequency ("RF"), etc., or any suitable combination of the foregoing.
(84) Computer program code for carrying out operations for aspects of the present disclosure may be written in any combination of one or more programming languages, including an object oriented programming language such as Java, Scala, Smalltalk, Eiffel, JADE, Emerald, C++, C#, VB.NET, Python or the like, conventional procedural programming languages, such as the "C" programming language, Visual Basic, Fortran 2003, Perl, Common Business Oriented Language ("COBOL") 2002, PHP: Hypertext Processor ("PHP"), Advanced Business Application Programming

("ABAP"), dynamic programming languages such as Python, Ruby and Groovy, or other programming languages. The program code may execute entirely on the user's computer, partly on the user's computer, as a stand-alone software package, partly on the user's computer and partly on a remote computer or entirely on the remote computer or server. In the latter scenario, the remote computer may be connected to the user's computer through any type of network, including a local area network (LAN) or a wide area network (WAN), or the connection may be made to an external computer (for example, through the Internet using an Internet Service Provider) or in a cloud computing environment or offered as a service such as a Software as a Service (Saas).
(85) Various embodiments were described herein with reference to flowchart illustrations and/or block diagrams of methods, apparatus (systems), devices and computer program products according to various embodiments described herein. It will be understood that each block of the flowchart illustrations and/or block diagrams, and combinations of blocks in the flowchart illustrations and/or block diagrams, can be implemented by computer program instructions. These computer program instructions may be provided to a processing circuit of a general purpose computer, special purpose computer, or other programmable data processing apparatus to produce a machine, such that the instructions, which execute via the processing circuit of the computer or other programmable instruction execution apparatus, create a mechanism for implementing the functions/acts specified in the flowchart and/or block diagram block or blocks.
(86) These computer program instructions may also be stored in a computer readable medium that when executed can direct a computer, other programmable data processing apparatus, or other devices to function in a particular manner, such that the instructions when stored in the computer readable medium produce an article of manufacture including instructions which when executed, cause a computer to implement the function/act specified in the flowchart and/or block diagram block or blocks. The computer program instructions may also be loaded onto a computer, other programmable instruction execution apparatus, or other devices to cause a series of operations to be performed on the computer, other programmable apparatuses or other devices to produce a computer implemented process such that the instructions which execute on the computer or other programmable apparatus provide processes for implementing the functions/acts specified in the flowchart and/or block diagram block or blocks.
(87) The flowchart and block diagrams in the figures illustrate the architecture, functionality, and operation of possible implementations of systems, methods, and computer program products according to various aspects of the present disclosure. In this regard, each block in the flowchart or block diagrams may represent a module, segment, or portion of code, which includes one or more executable instructions for implementing the specified logical function(s). It should also be noted that, in some alternative implementations, the functions noted in the block may occur out of the order noted in the figures. For example, two blocks shown in succession may, in fact, be executed substantially concurrently, or the blocks may sometimes be executed in the reverse order, depending upon the functionality involved. It will also be noted that each block of the block diagrams and/or flowchart illustration, and combinations of blocks in the block diagrams and/or flowchart illustration, can be implemented by special purpose hardware-based systems that perform the specified functions or acts, or combinations of special purpose hardware and computer instructions.
(88) The terminology used herein is for the purpose of describing particular aspects only and is not intended to be limiting of the disclosure. As used herein, the singular forms "a", "an" and "the" are intended to include the plural forms as well, unless the context clearly indicates otherwise. It will be further understood that the terms "comprises" and/or "comprising," when used in this specification, specify the presence of stated features, steps, operations, elements, and/or components, but do not preclude the presence or addition of one or more other features, steps, operations, elements, components, and/or groups thereof. As used herein, the term "and/or" includes any and all combinations of one or more of the associated listed items and may be

designated as “/”. Like reference numbers signify like elements throughout the description of the figures.
(89) Many different embodiments have been disclosed herein, in connection with the above description and the drawings. It will be understood that it would be unduly repetitious and obfuscating to literally describe and illustrate every combination and subcombination of these embodiments. Accordingly, all embodiments can be combined in any way and/or combination, and the present specification, including the drawings, shall be construed to constitute a complete written description of all combinations and subcombinations of the embodiments described herein, and of the manner and process of making and using them, and shall support claims to any such combination or subcombination.

Claims

1. A gaming device comprising: a housing defining an interior volume; a physical gameplay element disposed in the interior volume; a transparent display disposed on the housing, the transparent display comprising: an emissive layer comprising an array of emissive pixels, each emissive pixel of the array of emissive pixels comprising an emissive pixel element and a transparent portion; and a conductive layer operatively coupled to the emissive layer to selectively operate the emissive pixel elements; a processor circuit; and a memory storing machine-readable instructions that, when executed by the processor circuit, cause the processor circuit to selectively operate the transparent display, wherein the transparent portion of each emissive pixel is transparent to visible light in an inactive state such the physical gameplay element in the interior volume of the housing is visible through the transparent portion to a user outside the housing, and wherein each emissive pixel of the array of emissive pixels emits light in an active state sufficient to obscure visibility of the physical gameplay element in the interior volume of the housing through the transparent portion of the emissive pixel to the user outside the housing.
2. The gaming device of claim 1, wherein selective operation of the transparent display causes a first subset of the array of emissive pixels to be in the active state and a second subset of the array of emissive pixels to be in the inactive state.
3. The gaming device of claim 1, wherein the transparent display further comprises a transparent glass layer, wherein the emissive layer and the conductive layer are attached to the transparent glass layer.
4. The gaming device of claim 1, wherein the transparent display comprises a transparent Organic Light-Emitting Diode (OLED) display, and wherein the emissive pixel element of each emissive pixel comprises an OLED element.
5. The gaming device of claim 1, further comprising an interior display device disposed in the interior volume of the housing, wherein the interior display device is visible to the user outside the housing through the plurality of emissive pixels when the emissive pixels are in the inactive state.
6. The gaming device of claim 1, wherein the physical gameplay element comprises a plurality of mechanical slot reels disposed in the interior volume of the housing, wherein the plurality of mechanical slot reels is visible to the user outside the housing through the plurality of emissive pixels when the emissive pixels are in the inactive state.
7. The gaming device of claim 6, wherein the plurality of mechanical reels comprise a first curvature, and wherein a portion of the transparent display comprises a second curvature corresponding to the first curvature, wherein a distance between the portion of the transparent display and the plurality of mechanical reels is substantially uniform.
8. The gaming device of claim 1, wherein the physical gameplay element comprises a plurality of physical dice disposed in the interior volume of the housing, wherein the plurality of physical dice is visible to the user outside the housing through the plurality of emissive pixels when the emissive pixels are in the inactive state.

9. The gaming device of claim 1, wherein the physical gameplay element comprises a plurality of physical playing cards disposed in the interior volume of the housing, wherein the plurality of physical playing cards is visible to the user outside the housing through the plurality of emissive pixels when the emissive pixels are in the inactive state.
10. The gaming device of claim 1, further comprising a non-monetary prize disposed in the interior volume of the housing, wherein the non-monetary prize is visible to the user outside the housing through the plurality of emissive pixels when the emissive pixels are in the inactive state.
11. The gaming device of claim 1, wherein an area of the transparent portion of each emissive pixel comprises between 5% and 95% of a total area of the emissive pixel.
12. The gaming device of claim 1, wherein the array of emissive pixels comprises a pixel density of at least 50 dots per inch (DPI).
13. A gaming device comprising: a processor circuit; and a memory storing machine-readable instructions that, when executed by the processor circuit, cause the processor circuit to selectively operate a transparent display of a gaming device, the transparent display comprising: an emissive layer comprising an array of emissive pixels, each emissive pixel of the array of emissive pixels comprising an emissive pixel element and a transparent portion; and a conductive layer operatively coupled to the emissive layer to selectively operate the emissive pixel elements, wherein the transparent portion of each emissive pixel is transparent to visible light in an inactive state such a refillable gaming device component in an interior volume of the gaming device housing is visible through the transparent portion to a user outside the housing, and wherein each emissive pixel of the array of emissive pixels emits light in an active state sufficient to obscure visibility of the refillable gaming device component in the interior volume of the housing through the transparent portion of the emissive pixel to the user outside the housing.
14. The gaming device of claim 13, wherein selective operation of the transparent display causes a first subset of the array of emissive pixels to be in the active state and a second subset of the array of emissive pixels to be in the inactive state.
15. The gaming device of claim 13, wherein the refillable gaming device component comprises a ticket printer device comprising a paper receptacle, wherein a paper fill level of the paper receptacle is visible to the user outside the housing through the plurality of emissive pixels when the emissive pixels are in the inactive state.
16. The gaming device of claim 15, wherein a portion of the paper receptacle is transparent such that the paper fill level of the paper receptacle is visible to the user through the portion of the paper receptacle.
17. The gaming device of claim 13, wherein the refillable gaming device component comprises a token hopper device comprising a token receptacle, wherein a token fill level of the token receptacle is visible to the user outside the housing through the plurality of emissive pixels when the emissive pixels are in the inactive state.
18. The gaming device of claim 17, wherein a portion of the token receptacle is transparent such that the token fill level of the token receptacle is visible to the user through the portion of the token receptacle.
19. A gaming device comprising a transparent display, the transparent display comprising: an emissive layer comprising an array of emissive pixels, each emissive pixel of the array of emissive pixels comprising an emissive pixel element and a transparent portion; and a conductive layer operatively coupled to the emissive layer to selectively operate the emissive pixel elements; and a non-monetary prize receptacle disposed in the interior volume of the housing, wherein the transparent portion of each emissive pixel is transparent to visible light in an inactive state such an interior volume of a housing of the gaming device is visible through the transparent portion to a user outside the housing, wherein each emissive pixel of the array of emissive pixels emits light in an active state sufficient to obscure visibility of the interior volume of the housing through the transparent portion of the emissive pixel to the user outside the housing, and wherein, when anon-

monetary prize is disposed in the non-monetary prize receptacle, the non-monetary prize is visible to the user outside the housing through the plurality of emissive pixels when the emissive pixels are in the inactive state.

20. The gaming device of claim 19, further comprising a transparent glass layer, wherein the emissive layer and the conductive layer are attached to the transparent glass layer.